

Differential Equations

(first order & first degree).

- variables are separable

Solution is

$$f(x) dx = \varphi(y) dy$$

$$\int f(x) dx = \int \varphi(y) dy + C$$

- Homogeneous Equation

Homogeneous function :- A homogeneous function is a function $f(x_1, x_2, \dots, x_n)$ of several variables such that

$$f(cx_1, cx_2, \dots, cx_n) = c^k f(x_1, x_2, \dots, x_n)$$

for any $c \neq 0$, for all $x_1, x_2, \dots, x_n$. degree of homogeneity

Example

① $f(x, y) = x^2 + y^2 + xy \rightarrow$ Homogeneous of degree 2.

② $f(x, y) = x^2 + y^2 + xy \rightarrow$ Not Homogeneous

Homogeneous Equation :- $\frac{dy}{dx} = \frac{f(x, y)}{\varphi(x, y)}$

where both $f(x, y)$ & $\varphi(x, y)$ are homogeneous functions of same degree

To solve Homogeneous Equation :-

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\frac{dy}{dx} = \frac{f(x, y)}{\varphi(x, y)} \Rightarrow v + x \frac{dv}{dx} = \frac{x^k f(v)}{x^k \varphi(v)}$$

Example :-

Solve $(x^2 - y^2) dx - xy dy = 0$

$$\frac{(x^2 - y^2)}{xy} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{f(x, y)}{\varphi(x, y)}$$

$$\frac{(x^2 - y^2)}{xy} = \frac{dy}{dx}$$

$$\left(\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)} \right)$$

Put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\frac{x^2 - y^2}{xy} = \frac{x^2 - v^2 x^2}{x(vx)} = \frac{1 - v^2}{v}$$

$$\frac{1 - v^2}{v} = v + x \frac{dv}{dx}$$

$$\frac{1 - v^2}{v} - v = x \frac{dv}{dx}$$

$$\frac{1 - v^2 - v^2}{v} = x \frac{dv}{dx}$$

$$\int \frac{dx}{x} = \int \frac{v}{1 - 2v^2} dv$$

$$\ln(x) + C = \int -\frac{1}{4} \frac{dz}{z}$$

$$-\frac{1}{4} \ln(z) = \ln(x) + C$$

$$\ln(1 - 2v^2) = -4 \ln x + C'$$

$$\ln(1 - 2v^2) = \ln(x^{-4}) + \ln C''$$

$$1 - 2v^2 = \frac{C''}{x^4}$$

$$\dots \dots \dots$$

$$\begin{aligned} 1 - 2v^2 &= z \\ -4v dv &= dz \\ v dz &= -\frac{1}{4} dz \end{aligned}$$

$$C' = -4C$$

$$v = \frac{y}{x}$$

x^7

$$1 - \frac{2y^2}{x^2} = \frac{C''}{x^4}$$

$$x^4 - 2x^2y^2 = C''$$

* Differential Equations Reducible to Homogeneous Equation:-

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$$

$$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$$

$$x = X + h \quad y = Y + k, \quad h \text{ and } k \text{ are constants.}$$

$$a_1x + b_1y + c_1 = a_1(X + h) + b_1(Y + k) + c_1 \\ = a_1X + b_1Y + (a_1h + b_1k + c_1)$$

$$a_2x + b_2y + c_2 = a_2X + b_2Y + (a_2h + b_2k + c_2)$$

We have to choose h , and k so that

$$a_1h + b_1k + c_1 = 0 \\ a_2h + b_2k + c_2 = 0$$

$$\Rightarrow \begin{aligned} a_1h + b_1k &= -c_1 \\ a_2h + b_2k &= -c_2 \end{aligned}$$

$$\begin{aligned} AX &= B \\ X &= A^{-1}B \end{aligned}$$

$$a_1b_2 - a_2b_1 = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 0$$

No solution
OR Infinitely many solution

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0$$

Unique solution

$$a_1b_2 - a_2b_1 \neq 0$$

$$a_1 b_2 - a_2 b_1 = 0$$

$$a_1 b_2 = a_2 b_1$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

Working Rule

$$\frac{dy}{dx} = \frac{a_1 x + b_1 y + c_1}{a_2 x + b_2 y + c_2}$$

$$a_1 b_2 - a_2 b_1 = 0$$

↓

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{dy}{dx} = \frac{a_1 x + b_1 y + c_1}{m(a_1 x + b_1 y) + c_2}$$

↓

substitute $a_1 x + b_1 y = 0$

question :- solve

$$\frac{dy}{dx} = \frac{x + 2y - 3}{2x + y - 3}$$

Here

$$a_1 b_2 - b_1 a_2 = 1 \cdot 1 - 2(2) = -3 \neq 0$$

substitute

$$x = X + h$$

$$y = Y + k$$

$$\frac{dY}{dX} = \frac{(X+h) + 2(Y+k) - 3}{2(X+h) + (Y+k) - 3} = \frac{X + 2Y + (h + 2k - 3)}{2X + Y + (2h + k - 3)}$$

Find values of h and k so that

$$h + 2k = 3$$

$$\boxed{h = 1, \quad k = 1}$$

$$\begin{aligned} x &= X + h = X + 1 \\ Y &= Y - 1 \\ Y &= Y - 1 \end{aligned}$$

$$\begin{aligned} h+2k &= 3 \\ 2h+k &= 3 \end{aligned}$$

$$\boxed{h=1, \quad k=1}$$

$$\begin{aligned} x &= x-1 \\ y &= y-1 \end{aligned}$$

$$\frac{dy}{dx} = \frac{x+2y}{2x+y}$$

$$y = v x$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{x + 2vx}{2x + vx} = \frac{1+2v}{2+v}$$

$$x \frac{dv}{dx} = \frac{1+2v}{2+v} - v = \frac{1+2v-2v-v^2}{2+v}$$

$$\int \frac{2+v}{1-v^2} dv = \int \frac{dx}{x}$$

$$\int \frac{2+v}{(1+v)(1-v)} dv = \ln(x) + C$$

$$\begin{aligned} \int \frac{1}{2(1+v)} dv + \frac{3}{2} \int \frac{dv}{1-v} \\ = \ln(x) + C \end{aligned}$$

$$\frac{1}{2} \ln(1+v) - \frac{3}{2} \ln|1-v| = \ln(x) + C$$

$$\ln \left(\frac{1+v}{(1-v)^3} \right) = \ln x^2 + \ln C'$$

$$\frac{1+v}{(1-v)^3} = x^2 C'$$

$$v = \frac{y}{x}$$

$$\underline{x^2(x+y)} = x^2 C' \Rightarrow x+y = C' (x-y)^3$$

$$\begin{aligned} \frac{A}{1+v} + \frac{B}{1-v} &= \frac{2+v}{(1+v)(1-v)} \\ \frac{A(1-v) + B(1+v)}{(1+v)(1-v)} &= \frac{2+v}{(1+v)(1-v)} \end{aligned}$$

$$(A+B) + (B-A)v = 2+v$$

$$\begin{aligned} B-A &= 1 \\ A+B &= 2 \end{aligned}$$

$$\begin{aligned} 2B &= 3 \\ B &= 3/2 \end{aligned}$$

$$A = \frac{1}{2}$$

$$\frac{x^2(x+y)}{(x-y)^3} = x^2 c' \Rightarrow x+y = c' (x-y)^{-2}$$

$$\Rightarrow (x+y-2) = c' (x-y)^3.$$

Solve

$$\frac{dy}{dx} = \frac{3x-4y-2}{3x-4y-3}$$

Here

$$a_1 b_2 - b_1 a_2 = (-3, 4) - (-3, 4) = 0$$

Substitute

$$3x-4y = v$$

$$3 - 4 \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dy}{dx} = -\frac{1}{4} \left(\frac{dv}{dx} - 3 \right)$$

$$-\frac{1}{4} \left(\frac{dv}{dx} - 3 \right) = \frac{v-2}{v-3}$$

$$-\frac{1}{4} \frac{dv}{dx} + \frac{3}{4} = \frac{v-2}{v-3}$$

$$-\frac{1}{4} \frac{dv}{dx} = \frac{v-2}{v-3} - \frac{3}{4} = \frac{4(v-2)-3(v-3)}{4(v-3)} = \frac{v+1}{4(v-3)}$$

$$-\frac{dv}{dx} = \frac{v+1}{(v-3)}$$

$$-\frac{(v-3)}{v+1} dv = dx$$

$$\frac{v-3}{v+1} dv = -1 dx$$

$$x = x+h$$

$$y = y+k$$

$$3h-4k-2=0$$

$$3h-4k-3=0$$

$$\rightarrow 3h-4k=2$$

$$\rightarrow 3h-4k=3$$

$$\int \left(\frac{(v+1)-4}{v+1} \right) dv = \int -1 dx$$

$$v = 3x - 4y$$

$$v - 4 \ln(v+1) = -x + C$$

$$3x - 4y - 4 \ln(3x - 4y + 1) = -x + C$$

$$4x - 4y - 4 \ln(3x - 4y + 1) = C.$$

Question:- $x \frac{dy}{dx} = \frac{y^2}{x} + y \quad (x \neq 0)$

$$\frac{dy}{dx} = \frac{y^2}{x^2} + \frac{y}{x}$$

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = v^2 + v$$

$$x \frac{dv}{dx} = v^2$$

$$\Rightarrow \int \frac{dv}{v^2} = \int \frac{dx}{x}$$

$$\ln|x| + \frac{x}{y} + C = 0$$

$$-\frac{1}{v} = \ln|x| + C$$

$\Leftrightarrow$

$$-\frac{x}{y} = \ln|x| + C$$